FeelUp: AI Powered Emotion Sharing and Support Platform

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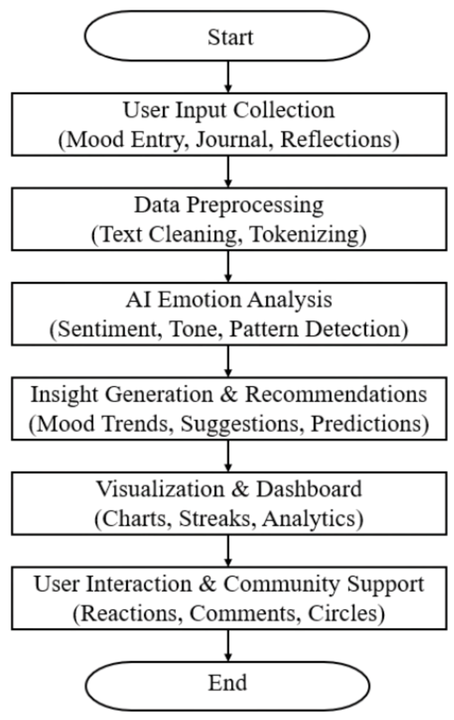
INTRODUCTION

Emotional well-being plays a crucial role in maintaining mental balance, healthy relationships, and overall quality of life. However, many individuals struggle to express and understand their emotions in the modern digital environment. Moreover, conventional social media platforms often prioritize content display and social comparison rather than emotional support. This research introduces FeelUp, an AI-powered platform designed to provide a secure, semi-anonymous emotional support environment where users can share experiences, track mood patterns, and receive peer-based emotional support. The platform integrates AI-driven emotion analysis, sentiment understanding, and behavioral analytics to provide meaningful insights and encourage emotional self-reflection. The methodology of the proposed system involves collecting user-generated emotional content through mood posts, journal entries, and community interactions, which represent natural language expressions of personal feelings and experiences. Before analysis, the system performs preprocessing steps including text normalization, removal of unsupported characters, input length validation, and anonymization of user identifiers to ensure privacy and consistent data handling. After preprocessing, a transformer-based embedding model converts the input text into semantic vector representations that capture contextual meaning. To identify emotionally relevant peer experiences, the system applies semantic retrieval techniques such as K-Nearest Neighbor (KNN) search, cosine similarity comparison, and HNSW indexing for efficient vector search and the system retrieves the most emotionally similar posts above a similarity threshold, enabling meaningful peer connections and supportive community interactions.

FEATURES AND IMPLEMENTATION

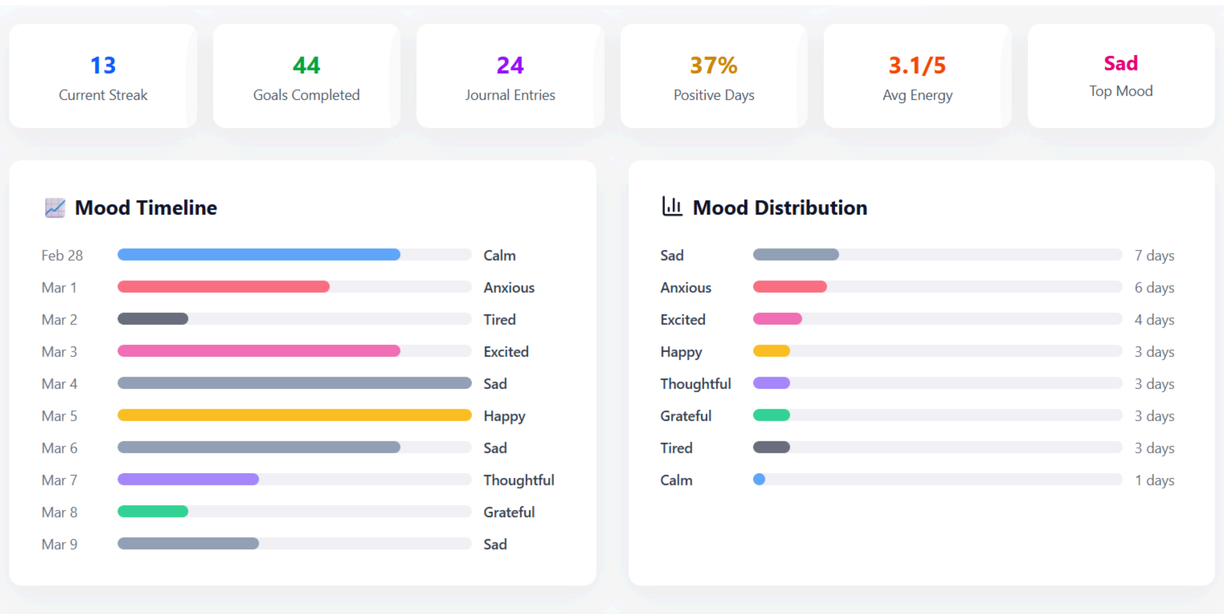
The FeelUp platform provides several intelligent features designed to support emotional well-being and meaningful community interaction. The system allows users to share their emotions through mood posts and journal entries, enabling a safe and semi-anonymous environment for emotional expression. It incorporates Natural Language Processing (NLP) techniques to analyze user-generated text and understand emotional context. Based on semantic similarity, the platform connects users who are experiencing similar emotional situations, promoting peer support and empathetic communication.



Metric	Value
Total Queries	30
Zero-hit Queries	4 (13.3 %)
Precision@5	0.8667
Average Cosine Similarity	0.6890
Cohen's Kappa (κ)	1.0

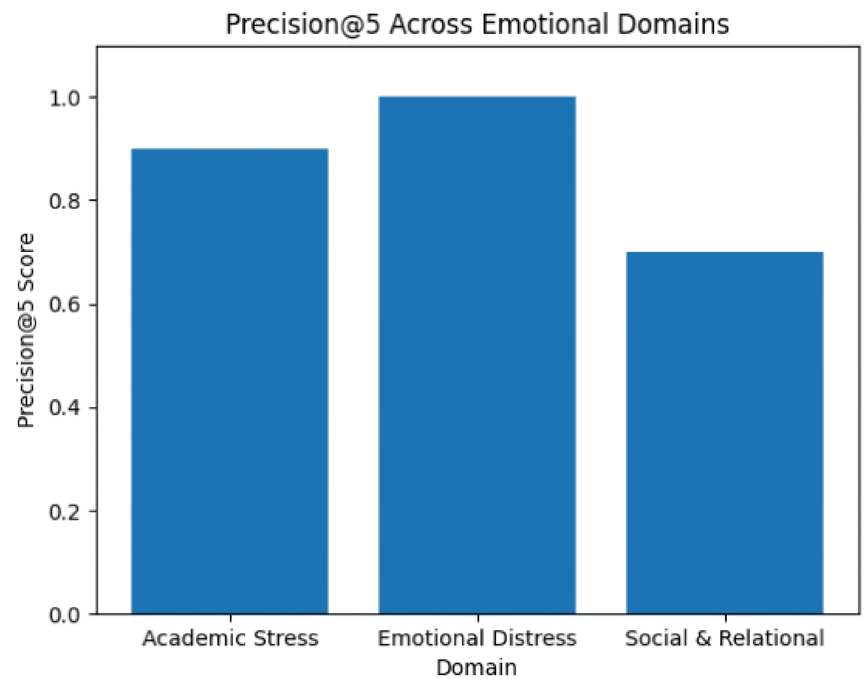
RESULTS AND DISCUSSION

The FeelUp platform was evaluated to measure the effectiveness of its semantic peer-matching system and overall performance in retrieving emotionally relevant content. The evaluation was conducted using a set of emotional queries representing different contexts such as academic stress, emotional distress, and social or relational situations. For each query, the system retrieved the most semantically similar posts using transformer-based embeddings and Approximate Nearest Neighbor (ANN) vector search. The results showed strong performance, with the majority of retrieved posts being emotionally relevant to the input query, indicating that the system successfully captures contextual emotional meaning rather than relying on simple keyword matching. The average semantic similarity scores demonstrated consistent alignment between user queries and retrieved peer experiences.



Additionally, the system includes mood tracking and analytics features that help users monitor emotional patterns over time and gain insights into their mental well-being. A community interaction feed allows users to engage with supportive posts, while a visualization dashboard presents mood trends, journaling activity, and behavioral insights to encourage emotional self-reflection.

Domain	Precision @5	Avg. Similarity
Academic Stress	0.9000	0.6803
Emotional Distress	1.0000	0.6966
Social & Relational	0.7000	0.6858



CONCLUSIONS

The FeelUp platform demonstrates how Artificial Intelligence can be effectively integrated into digital environments to support emotional well-being and meaningful social interactions. By utilizing transformer-based sentence embeddings and Approximate Nearest Neighbor (ANN) vector search, the system enables semantic understanding of user-generated emotional content and facilitates connections between individuals experiencing similar emotional situations. The evaluation results show strong retrieval performance, with high precision and semantic similarity scores, indicating that the platform can accurately identify emotionally relevant peer experiences. Additionally, the architecture maintains real-time responsiveness while preserving user privacy through secure data handling mechanisms. Overall, the proposed system highlights the potential of AI-driven technologies to enhance emotional awareness, encourage supportive community engagement, and provide a scalable foundation for future digital mental wellness platforms.